

A positive witness—or a complete obstruction atlas

For prime $p \equiv 1 \pmod{4}$, retain every shell $p/4 < a < 3p/4$, every actual divisor $u \mid a^2$, and both channels. Put $R = 4a - p$.

Every candidate already gives a *positive rational* decomposition of $4/p$. The last two denominators have the same exact reduced denominator:

$$\boxed{D_E = \frac{R}{\gcd(R, 4u + 1)}, \quad D_M = \frac{R}{\gcd(R, u + a)}}.$$

Integrality at the original prime is exactly $D = 1$.

Constructive direction

An actual signed divisor box has multiplicity mass N and collision energy $E = \sum_g \mu(g)^2$. A miss in M allowed residue bins forces

$$E \geq Mq^2 + r(2q + 1), \quad N = Mq + r, \quad 0 \leq r < M.$$

At $p = 1201$, $a = 306$, $R = 23$:

$$N = 45, \quad M = 19, \quad E = 107 < 111.$$

This forces the marked exterior witness

$$(R, u, h, r, s, \kappa) = (23, 17, 17, 1, 18, 53).$$

The ordinary Cauchy bound does not force this hit.

Counterfactual direction

If ES fails at p , the complete finite atlas has $D \geq 3$ everywhere. Define the labelled diagonal operator

$$H_p = \text{diag}\left(\frac{D-1}{2}\right).$$

It must have the exact positive inverse

$$H_p^{-1} = \text{diag}\left(\frac{2}{D-1}\right), \quad 0 < H_p^{-1} \leq I.$$

Every original exponent-box leaf must carry a certified prime-valuation obstruction. Restricted negative examples are not global counterexamples.

Clearing by D gives a genuine integral solution at pD , not at p . All factor labels, affine scales, Leech coordinates, and the full cubic correction remain attached.

Complete proofs and standard-library replay: `workbench.tex`, `dual_es.py`, and two three-page notes.

No global ES proof or counterexample is claimed. Exact refinements of inherited coordinates; historical priority unassessed.